Practical -15
Module Name: Selenium Web Automation
Test Case ID:P–15

- **Scenario Id: 1**
- **Name Ai**

Test step:

1. **Open the pycharm**
2. **Write a code.**
3. **Open the Terminal.**
4. **Click on run.**

- **Code:**

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import (
    NoSuchElementException,
    TimeoutException,
    ElementNotInteractableException,
    StaleElementReferenceException
)

URL = "https://www.saucedemo.com/"
VALID_USERNAME = "standard_user"
VALID_PASSWORD = "secret_sauce"


def start_browser():
    driver = webdriver.Chrome()
    driver.maximize_window()
    return driver


def login(driver, username, password):
```

```
    driver.get(URL)
    wait = WebDriverWait(driver, 10)

    username_box = wait.until(
        EC.visibility_of_element_located((By.ID, "user-name"))
    )

    password_box = wait.until(
        EC.visibility_of_element_located((By.ID, "password"))
    )

    login_button = wait.until(
        EC.element_to_be_clickable((By.ID, "login-button"))
    )

    username_box.clear()
    username_box.send_keys(username)

    password_box.clear()
    password_box.send_keys(password)

    login_button.click()


# TC-01: Valid Login
def test_valid_login():
    driver = start_browser()

    try:
        login(driver, VALID_USERNAME, VALID_PASSWORD)

        wait = WebDriverWait(driver, 10)

        wait.until(EC.url_contains("inventory"))

        assert "inventory" in driver.current_url

        products_title = wait.until(
            EC.visibility_of_element_located((By.CLASS_NAME, "title"))
        )

        assert products_title.text == "Products"

        print("TC-01 Valid Login: PASS")
```

```
        print("Authentication Assertion: PASS")

    except AssertionError:
        print("TC-01 Valid Login: FAIL")

    except NoSuchElementException:
        print("TC-01: NoSuchElementException handled")

    except TimeoutException:
        print("TC-01: TimeoutException handled")

    except ElementNotInteractableException:
        print("TC-01: ElementNotInteractableException handled")

    except StaleElementReferenceException:
        print("TC-01: StaleElementReferenceException handled")

    finally:
        driver.quit()


# TC-02: Invalid Password
def test_invalid_password():
    driver = start_browser()

    try:
        login(driver, VALID_USERNAME, "wrong_password")

        error_message = WebDriverWait(driver, 10).until(
            EC.visibility_of_element_located(
                (By.CSS_SELECTOR, "h3[data-test='error']")
            )
        )

        assert (
            "Username and password do not match"
            in error_message.text
        )

        print("TC-02 Invalid Password: PASS")

    except AssertionError:
        print("TC-02 Invalid Password: FAIL")
```

```
    except TimeoutException:
        print("TC-02: Error message was not displayed")

    finally:
        driver.quit()


# TC-03: Invalid Username
def test_invalid_username():
    driver = start_browser()

    try:
        login(driver, "wrong_user", VALID_PASSWORD)

        error_message = WebDriverWait(driver, 10).until(
            EC.visibility_of_element_located(
                (By.CSS_SELECTOR, "h3[data-test='error']")
            )
        )

        assert (
            "Username and password do not match"
            in error_message.text
        )

        print("TC-03 Invalid Username: PASS")

    except AssertionError:
        print("TC-03 Invalid Username: FAIL")

    except TimeoutException:
        print("TC-03: Error message was not displayed")

    finally:
        driver.quit()


# TC-04: Empty Credentials
def test_empty_credentials():
    driver = start_browser()

    try:
        driver.get(URL)
```

```
        login_button = WebDriverWait(driver, 10).until(
            EC.element_to_be_clickable((By.ID, "login-button"))
        )

        login_button.click()

        error_message = WebDriverWait(driver, 10).until(
            EC.visibility_of_element_located(
                (By.CSS_SELECTOR, "h3[data-test='error']")
            )
        )

        assert "Username is required" in error_message.text

        print("TC-04 Empty Credentials: PASS")

    except AssertionError:
        print("TC-04 Empty Credentials: FAIL")

    except TimeoutException:
        print("TC-04: Validation message was not displayed")

    finally:
        driver.quit()


# TC-05: Logout
def test_logout():
    driver = start_browser()

    try:
        login(driver, VALID_USERNAME, VALID_PASSWORD)

        wait = WebDriverWait(driver, 10)

        wait.until(EC.url_contains("inventory"))

        menu_button = wait.until(
            EC.element_to_be_clickable(
                (By.ID, "react-burger-menu-btn")
            )
        )

        menu_button.click()
```

```
        logout_button = wait.until(
            EC.element_to_be_clickable(
                (By.ID, "logout_sidebar_link")
            )
        )

        logout_button.click()

        wait.until(
            EC.visibility_of_element_located(
                (By.ID, "login-button")
            )
        )

        assert driver.current_url.rstrip("/") == URL.rstrip("/")

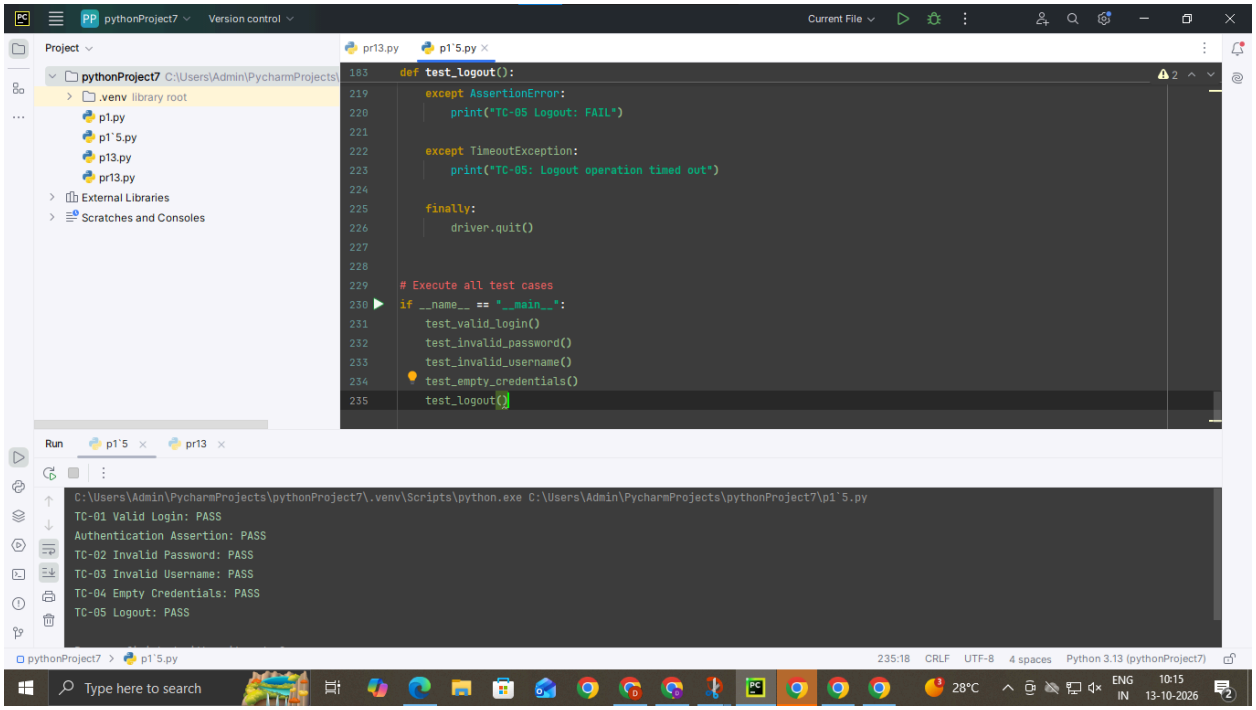
        print("TC-05 Logout: PASS")

    except AssertionError:
        print("TC-05 Logout: FAIL")

    except TimeoutException:
        print("TC-05: Logout operation timed out")

    finally:
        driver.quit()


# Execute all test cases
if __name__ == "__main__":
    test_valid_login()
    test_invalid_password()
    test_invalid_username()
    test_empty_credentials()
    test_logout()
```


pythonProject7
Version control
Current File
Project
pythonProject7 C:\Users\Admin\PycharmProjects\
.venv library root
p1.py
p1`5.py
p13.py
pr13.py
External Libraries
Scratches and Consoles
pr13.py
p1`5.py
def test_logout():
except AssertionError:
print("TC-05 Logout: FAIL")
except TimeoutException:
print("TC-05: Logout operation timed out")
finally:
driver.quit()
Execute all test cases
if __name__ == "__main__":
test_valid_login()
test_invalid_password()
test_invalid_username()
test_empty_credentials()
test_logout()
Run
p1`5
pr13
C:\Users\Admin\PycharmProjects\pythonProject7\.venv\Scripts\python.exe C:\Users\Admin\PycharmProjects\pythonProject7\p1`5.py
TC-01 Valid Login: PASS
Authentication Assertion: PASS
TC-02 Invalid Password: PASS
TC-03 Invalid Username: PASS
TC-04 Empty Credentials: PASS
TC-05 Logout: PASS
235:18
CRLF
UTF-8
4 spaces
Python 3.13 (pythonProject7)
Type here to search
28°C
ENG
IN
10:15
13-10-2026